

STOR 538 Final Project Visuals

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Load/Clean Data

```
corners2 <- corners %>%
  mutate(new_location = ifelse(corners$Location == "Short" &
    corners$Left.Side == 1, "Short Left", ifelse(corners$Location == "Short" &
    corners$Left.Side == 0, "Short Right", Location)))

goal_locations <- data.frame(
  x = c(93, 91, 94, 99, 99, 100, 81, 95, 98, 90, 78, 93, 100, 96),
  y = c(50, 45, 58, 53, 47, 100, 40, 46, 55, 53, 29, 54, 50, 51)
)

zones <- data.frame(
  zone = c("Short Left", "Penalty Spot", "Back Post", "Goalmouth",
    "Front Post", "Other", "Top of Box", "Short Right"),
  xmin = c(80, 83, 83, 94, 83, 0, 75, 80),
  xmax = c(100, 94, 100, 101, 100, 0, 83, 100),
  ymin = c(0, 43, 57, 43, 22, 0, 23, 80),
  ymax = c(20, 57, 78, 57, 43, 0, 77, 100)
)

location_count_corners <- corners2 %>%
  count(new_location) %>%
  rename(zone = new_location)

data_w_zonecount <- zones %>%
  left_join(location_count_corners, by = "zone")

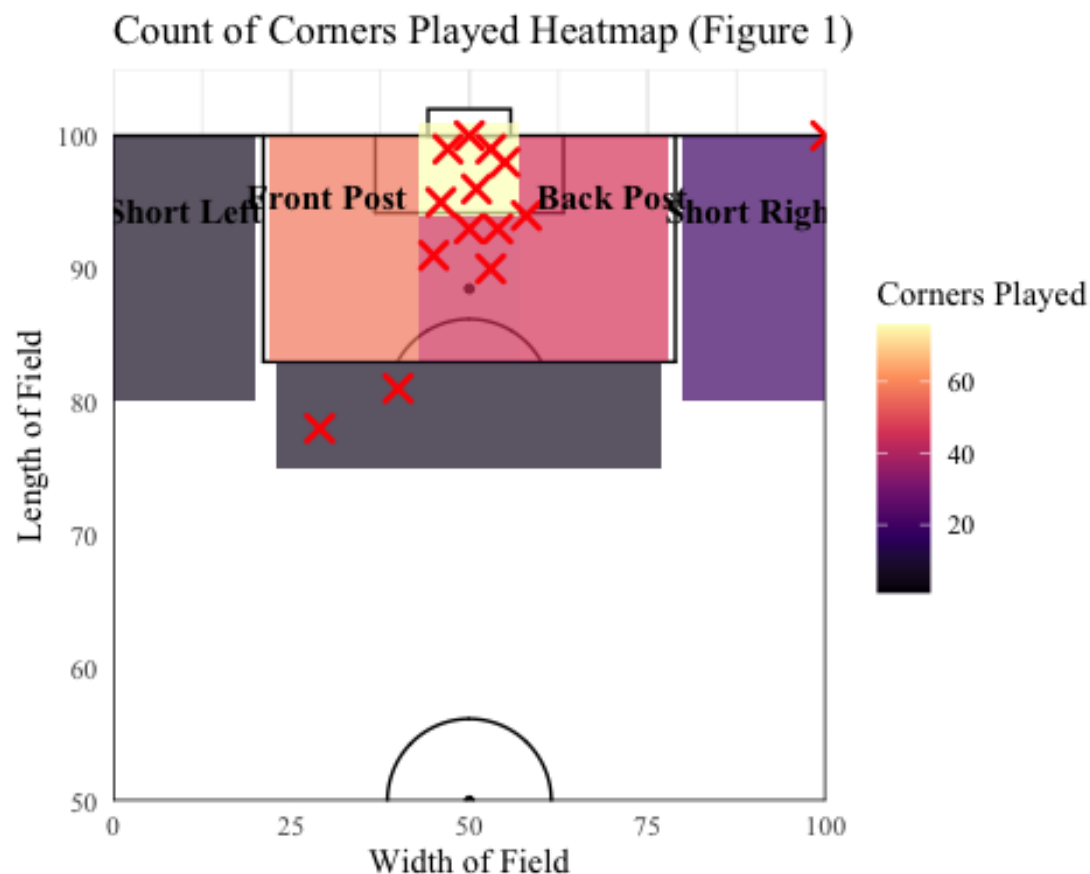
plot1 <- ggplot(data_w_zonecount) +
  annotate_pitch(colour = "black", fill = "white") +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = ymin, ymax = ymax,
    fill = n), alpha = 0.7) +
  geom_point(data = goal_locations, aes(x, y),
    shape = 4, size = 3, stroke = 1.3, color = "red") +
  coord_flip(xlim = c(50,105), ylim = c(0,100), expand = FALSE) +
  scale_fill_viridis_c(option = "magma", name = "Corners Played") +
  labs(title = "Count of Corners Played Heatmap (Figure 1)",
    x = "Length of Field",
    y = "Width of Field",
    family = "Times New Roman") +
  geom_text(data = data.frame(x = c(90, 91, 91, 90),
```

```

y = c(10, 30, 70, 90),
label = c("Short Left", "Front Post", "Back
Post", "Short Right")),
  vjust = -2, size = 4, family = "Times New Roman", fontface =
"bold", aes(x, y, label = label))+
  theme_minimal(base_family = "Times New Roman")

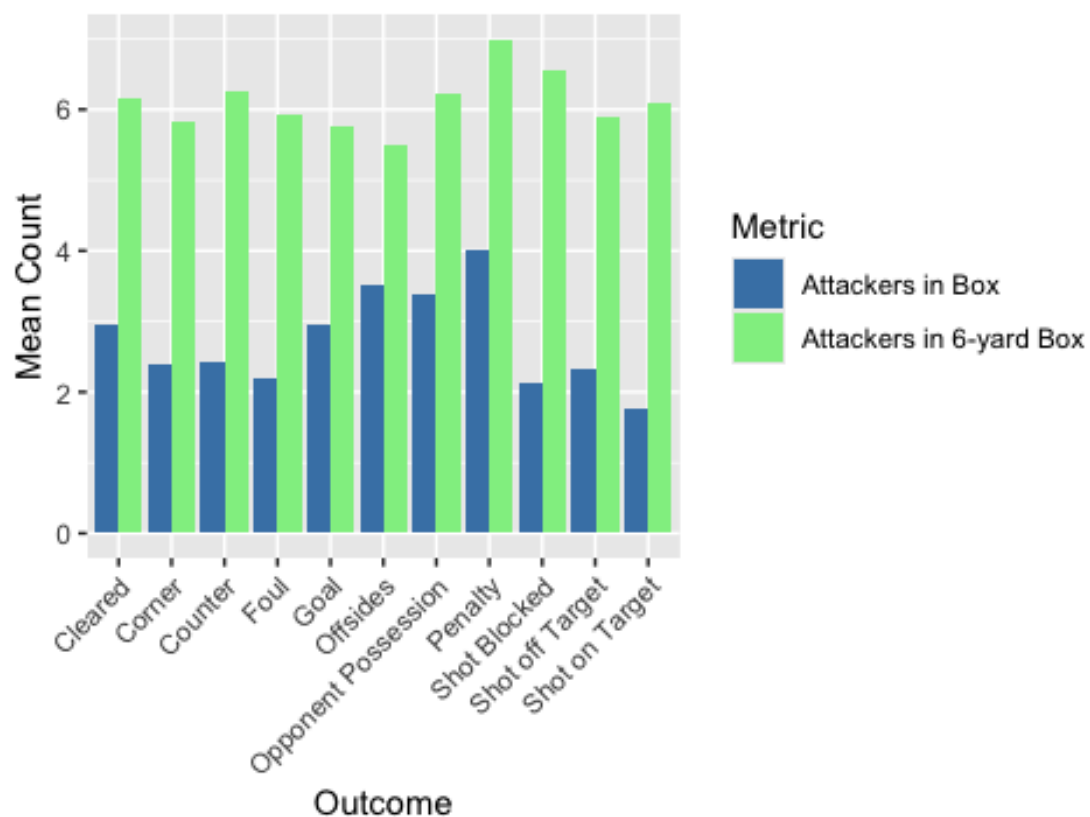
```

plot1



This heat map (Figure 1) was created to illustrate the number of corners played into at each location on the field, as well as the players' locations for each recorded goal. Each zone represents an area of the field recorded in our "Location" variable. Short passes were split based on whether they occurred on the right or left side of the field. In this figure, the left side of the field and the front post are shown on the left half, while the right side of the field and the back post are shown on the right half. It's important to note that we did not split front and back posts by side of the field, and the counts you see are the totals for front/back posts, not counting the side of the field they were on. Each red X-mark represents the location of a goalscorer from where he kicked the ball into the goal. For example, one goal was scored from the corner kick itself thus, the X-mark is in the corner from which he scored.

Average of Attackers for Each Outcome (Figure 2)



This bar plot(Figure 2) was created to display the average number of attackers in the box (A.in.Box) and in the 6-yard box (A.in.6) for each different outcome. Each outcome has two bars: the blue bar, representing the mean count of attackers in the box for each possible outcome, and the green bar, representing the mean count of attackers in the 6-yard box.

```
freq_table <- corners %>%
  count(Location, Outcome) %>%
  pivot_wider(names_from = Outcome,
              values_from = n,
              values_fill = 0)
```

```
freq_table
```

```
## # A tibble: 7 × 12
##   Location      Cleared Corner Counter  Foul  Goal `Opponent Possession`
##   <chr>          <int>  <int>  <int> <int> <int>          <int>
## 1 Back Post      15      4      3     1     3           9
## 2 Front Post    33      4      3     3     0           5
## 3 Goalmouth     24      3      3     7     6          13
```

```

0
## 4 Other          0      0      0      0      0      1
0
## 5 Penalty Spot   16      4      3      3      1      6
0
## 6 Short          4       2      0      1      5      4
0
## 7 Top of Box     2       1      0      0      1      0
0
## # i 4 more variables: `Shot Blocked` <int>, `Shot off Target` <int>,
## #   `Shot on Target` <int>, Offsides <int>

df_prop <- corners %>%
  mutate(LocationGroup = ifelse(Location == "Short", "Short", "Other")) %>%
  count(LocationGroup, Outcome) %>%
  group_by(LocationGroup) %>%
  mutate(prop = n / sum(n))

ggplot(df_prop, aes(x = LocationGroup, y = prop, fill = Outcome)) +
  geom_col() +
  labs(title = "Outcome Proportions: Short vs Other Corners (Figure 2)",
       x = "Corner Type",
       y = "Proportion",) +
  scale_fill_viridis_d(option = "turbo", name = "Outcome")+
  scale_y_continuous(labels = scales::percent) +
  theme_minimal(base_family = "Times New Roman")

```

Outcome Proportions: Short vs Other Corners (Figure 2)

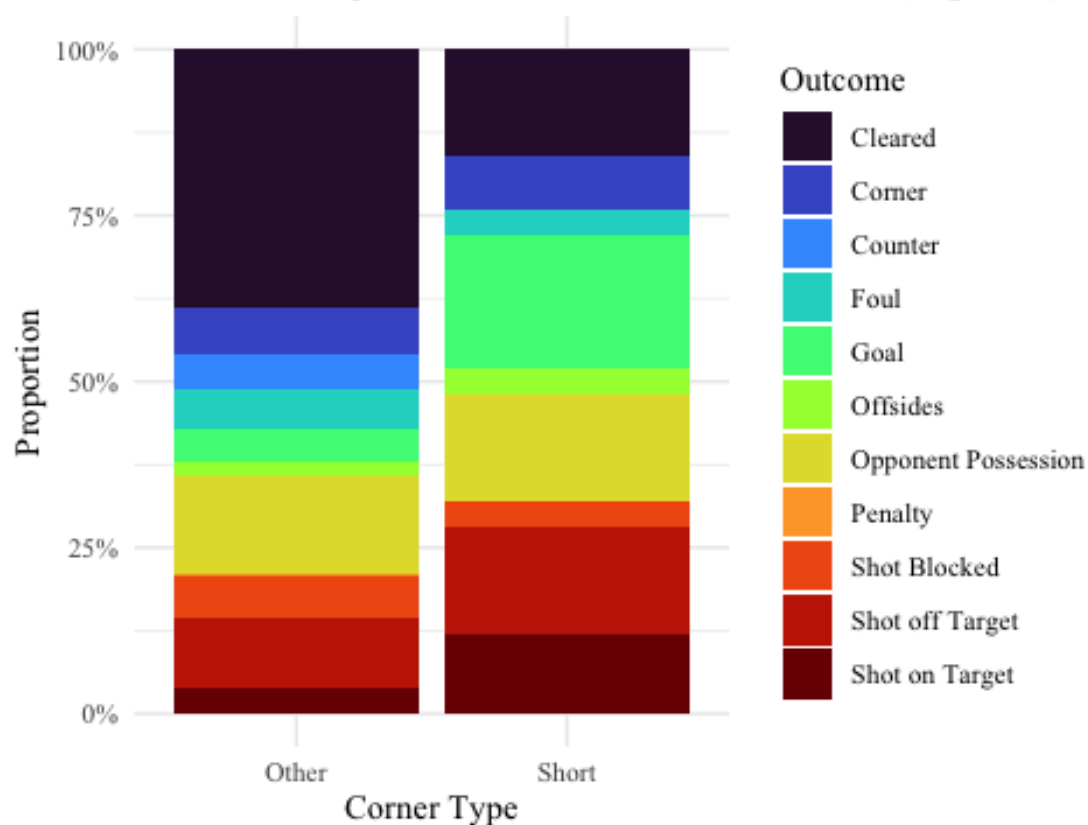


Figure 2 is a stacked bar chart illustrating the difference in proportions of outcomes for short corners compared to all other locations of corners. Each color of the bar represents the proportion of a specific outcome.

```
df_prop
## # A tibble: 20 × 4
## # Groups:   LocationGroup [2]
##   LocationGroup Outcome          n    prop
##   <chr>          <chr>      <int>  <dbl>
## 1 Other         Cleared          90 0.390
## 2 Other         Corner           16 0.0693
## 3 Other         Counter          12 0.0519
## 4 Other         Foul             14 0.0606
## 5 Other         Goal             11 0.0476
## 6 Other         Offsides           5 0.0216
## 7 Other         Opponent Possession 34 0.147
## 8 Other         Penalty           1 0.00433
## 9 Other         Shot Blocked      15 0.0649
## 10 Other        Shot off Target   24 0.104
## 11 Other        Shot on Target    9 0.0390
## 12 Short        Cleared           4 0.16
## 13 Short        Corner            2 0.08
```

## 14 Short	Foul	1 0.04
## 15 Short	Goal	5 0.2
## 16 Short	Offsides	1 0.04
## 17 Short	Opponent Possession	4 0.16
## 18 Short	Shot Blocked	1 0.04
## 19 Short	Shot off Target	4 0.16
## 20 Short	Shot on Target	3 0.12

```
write_csv(df_prop, "ShortVsOtherProps.csv")
```

Figure 2 showed some interesting proportions comparing short corners to all other corners. For example, 12% of short corners led to a shot on target while only 4% of all other corners led to shots on target. This led us to wonder whether or not short corners were more dangerous than the others.